

Recognise line and rotational symmetry of shapes and create symmetrical patterns and pictures, using dynamic geometric software where appropriate

Learning intention

- recognise line and rotational symmetry of shapes and create symmetrical patterns and pictures, using dynamic geometric software where appropriate.

Teach and model

- using stimulus materials such as the motifs in Central Asian textiles, Tibetan artefacts.
- identifying rotational symmetry of shapes by tracing around various shapes and objects to create an image.
- using dynamic geometric software or generative artificial intelligence tool to manipulate shapes and create symmetrical patterns.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.